

PREPARED FOR: TREVOR KNIGHT, GMD · FOR SEAN PARK / OED + MAYOR JOHN CARNEY DISCUSSION

ForwardTech Industrial Campus — Wilmington

Merge4Ward Manufacturing + M4CleanTech AD/RNG + Tire BOO · Hay Road Peninsula DE

A co-located clean manufacturing and clean tech campus on the Hay Road Peninsula, Wilmington DE — using a single City ground lease to simultaneously establish the Forward Building System's Mid-Atlantic CLT/SIP manufacturing node, an AD/RNG facility, and a Klean/KBR tire BOO. The ground lease provides site control that unlocks both facilities and a 12–22% blended reduction in Merge4Ward's per-panel material cost. Revenue-first.

Merge4Ward LLC · M4CleanTech LLC · Merge 4 Design LLC · M4Di™ · Aubyn Architecture LLC

SECTION 1 — FORWARDTECH CAMPUS REVENUE AT STEADY STATE

What the Campus Earns — All Three Components

The ForwardTech Industrial Campus generates revenue from three co-located components sharing one City ground lease. The ground lease is not just a real estate instrument — it is a procurement credential that unlocks a 12–22% blended material cost reduction for Merge4Ward, equivalent to \$600K–\$1.2M in additional annual gross margin. All figures ROM ±30–40% before operating expenses.

\$1.2M– \$1.6M Merge4Ward Manufacturing NOI Panel fabrication revenue minus material (with supply agreements) minus ops · Year 3	\$3.5M– \$6.9M M4CleanTech DE — AD/RNG Gross Tipping fees + D3 RINs + digestate · Year 3 steady state	\$150K– \$300K Tire BOO Ground Lease Spread Klean/KBR sublease net of City rent · Year 2 of BOO ops	\$4.85M– \$8.8M TOTAL CAMPUS GROSS · Annual All components at Year 3–5 steady state · Procurement savings embedded in MW NOI
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Campus Revenue Breakdown — All Streams

REVENUE / COST STREAM	DESCRIPTION	ROM AMOUNT	TIMING
Merge4Ward — Panel Sales Revenue	CLT/SIP/AAC panels sold to M4D project pipeline + third-party developers in	\$4.2M–\$7.8M gross	Year 2–3 ramp

REVENUE / COST STREAM	DESCRIPTION	ROM AMOUNT	TIMING
Merge4Ward — Material Cost (Market)	DE/PA/NJ/MD market. Year 3 at 40–60% facility utilization. Without ground lease / supplier agreements. Spot buyer pricing across all systems. No volume discount, no supply security.	(\$2.8M–\$5.6M)/yr	Baseline — no leverage
Merge4Ward — Material Cost (With Agreements)	12–22% blended reduction from confirmed ground lease procurement leverage + direct OEM supply agreements with CLT/SIP/AAC/sheathing/ICF suppliers.	(\$2.2M–\$4.4M)/yr	Post-agreement
Merge4Ward — Gross Margin Enhancement	Procurement savings flow directly to Merge4Ward gross margin. Equivalent to \$600K–\$1.2M annual revenue increase without selling additional panels.	+\$600K–\$1.2M/yr	Recurring annual
Merge4Ward NOI (est.)	Revenue minus material (with agreements) minus labor/ops/facility/debt service. Year 3 steady state.	\$1.2M–\$1.6M/yr	Year 3+
M4CleanTech DE — AD/RNG Tipping Fees	Port of Wilmington logistics corridor creates premium access to institutional and commercial food waste + FOG from northern DE + southern PA + MD Eastern Shore.	\$1.8M–\$3.2M/yr	Year 1 ops
M4CleanTech DE — D3 RIN Credits	Federal RFS D3 RIN credits for Wilmington food-waste-derived RNG. Same structure as Chester — 11.7 D3 RINs/MMBtu at \$2–\$4/gal market price.	\$1.4M–\$3.1M/yr	Year 1 ops
M4CleanTech DE — Digestate	Soil amendment sales to DE/MD regional agriculture. Barge access from Hay Road peninsula gives distribution cost advantage.	\$300K–\$600K/yr	Year 2+
Tire BOO — Ground Lease Spread	Klean/KBR sublease on southern peninsula parcel (26-052-00-004). Three-sided river frontage = barge offtake for pyrolytic oil + recovered carbon black. Zero M4CT capital.	\$150K–\$300K/yr	Year 2 BOO ops
CAMPUS TOTAL — Annual Recurring	Merge4Ward NOI + AD/RNG gross + Tire BOO spread. Procurement savings embedded in MW NOI above.	\$4.85M–\$8.8M/yr	Year 2–3+

REVENUE / COST STREAM	DESCRIPTION	ROM AMOUNT	TIMING
City of Wilmington — Tax Ratables + Jobs	Merge4Ward: 40–80 mfg. jobs. M4CleanTech DE: 25–40 clean tech ops jobs. New assessed ratables on brownfield peninsula currently generating minimal revenue.	65–120 jobs + \$27M–\$32M ratables	Year 2–4

ROM ±30–40%. Merge4Ward NOI assumes 12–22% material cost reduction from confirmed supply agreements. AD/RNG gross assumes 150,000–300,000 tons/yr organic feedstock. D3 RIN at \$2–\$4/gal stress-tested at \$1.50 floor. Tire BOO spread net of City rent.

SECTION 2 — THE SITE: HAY ROAD PENINSULA, WILMINGTON DE

12+ Acres. Three-Sided River Frontage. City Jurisdiction. Port Access.

The Hay Road Peninsula is a horseshoe-shaped industrial promontory where the Christina River wraps around three sides before meeting the Delaware River. More than 12 acres of available or underproductive parcels carrying 26-series NCC parcel IDs are confirmed visible on aerial imagery. The peninsula sits within City of Wilmington jurisdiction, inside the Port of Wilmington logistics corridor, and is currently underutilized. M4D has an existing relationship with Sean Park at City OED — this is a deepening conversation, not a cold ask.

MERGE4WARD MANUFACTURING	M4CLEANTECH DE — AD/RNG	TIRE BOO + SHARED INFRA
Parcels: 26-044-00-014 / -018 4–6 acres. Western cluster. Existing heavy truck access. Potential active CSX/NS rail spur — inbound CLT/SIP lumber, outbound panel distribution. 30,000–50,000 SF manufacturing + panel staging laydown yard. Target: 40–80 manufacturing jobs, \$4.2M–\$7.8M gross revenue/yr	Parcels: 26-044 / central cluster 3–6 acres. AD/RNG digesters + biogas upgrading + pipeline interconnection. Enbridge/SJI gas infrastructure accessible from peninsula. Barge offtake via river frontage for digestate. Food waste receives from Port of Wilmington logistics network. Target: \$3.5M–\$6.9M AD/RNG gross revenue/yr + 25–40 ops jobs	Parcel: 26-052-00-004 (southern tip) 3–5 acres. Three-sided river frontage — barge offtake for pyrolytic oil, recovered carbon black, and steel gives Wilmington tire BOO a logistics advantage over any inland TDP site. Klean/KBR builds and owns. Shared rail spur access with Merge4Ward. Target: \$150K–\$300K ground lease spread/yr · Zero M4CT CapEx
<i>Confirm: parcel ownership (City vs. private), rail spur status, Phase I ESA history</i>	<i>Confirm: pipeline interconnection feasibility with Enbridge/SJI before design commitment</i>	<i>ROFR on adjacent parcels must be in ground lease — protects Wilmington Phase 2 site</i>

SINGLE NEGOTIATION. THREE REVENUE STREAMS.

A single master development agreement with the City of Wilmington OED covers the entire peninsula — Merge4Ward manufacturing + M4CleanTech AD/RNG + Tire BOO + shared infrastructure. One ground lease conversation with Sean Park activates all three revenue streams. The City gets 65–120 permanent

jobs, brownfield activation, Opportunity Zone investment, and new tax ratables on a peninsula currently generating almost nothing. M4D gets site control at zero acquisition cost.

SECTION 3 — PROCUREMENT LEVERAGE: THE GROUND LEASE AS NEGOTIATING CREDENTIAL

Confirmed Site Control Changes Every Supplier Conversation by 12–22%

A signed long-term ground lease with the City of Wilmington transforms Merge4Ward from a startup buyer (priced with full uncertainty premium) to an anchor manufacturer with a committed, site-locked production ramp. Suppliers price uncertainty. When site uncertainty disappears, margin compression follows. The table below shows the before/after for each Merge4Ward material system.

SUPPLIER / SYSTEM	WITHOUT GROUND LEASE	WITH CONFIRMED GROUND LEASE	ROM SAVINGS	DELTA
CLT Panels (TerraLam / Structurlam / Mercer)	Spot buyer: \$180–\$240/m ³ . No volume discount. 12–16 week lead times. No priority.	Anchor manufacturer: \$145–\$195/m ³ at contracted annual volume. 6–8 wk lead. Supply priority. Potential vendor financing.	~\$35–\$55/m ³ savings	18–23%
SIP Panels (Acme / Premier / Panelwrights)	Project-by-project. Full list price. No standing order discount. Queue position uncertain.	Standing annual supply agreement: 8–15% volume discount. Dedicated fab queue. Co-branding as Mid-Atlantic distributor.	8–15% reduction on SIP cost	8–15%
AAC Block / Panel (Aercon / Xella / Brickform)	Standard distributor pricing + \$15–\$30/panel freight from nearest warehouse.	Direct mfr. supply at Wilmington node. DE river port freight savings + 10–18% volume pricing.	10–18% product + freight	10–18%
Thermo-Brace Structural Sheathing	Broker/distributor pricing. No direct OEM. No preferred access to limited supply runs.	Direct OEM agreement as regional anchor. Priority allocation + 10–15% below distributor.	10–15% below distributor + supply security	10–15%
Framecrete / ICF (Nudura / Fox Blocks / Logix)	Contractor pricing tier. No volume rebate. No regional distribution rights.	Regional distributor or preferred installer: 12–20% below contractor + DE/PA/NJ/MD exclusivity.	12–20% cost reduction + territorial moat	12–20%
AGGREGATE IMPACT (All systems, Yr 2–3)	Standard market pricing. No leverage. No supply security. Higher per-panel cost.	Blended material cost reduction: 12–22% across CLT/SIP/AAC/sheathing/ICF.	\$600K–\$1.2M/yr gross margin increase	12–22% blended

ROM cost delta estimates based on published manufacturer volume discount schedules, regional freight savings from waterfront DE location, and comparable preferred supplier agreements in the CLT/mass timber market. Actual savings depend on negotiated supplier agreement terms, committed annual volumes, and product mix. Merchant review required before committing to any supply agreement.

SECTION 4 — FORWARDTECH CAMPUS FINANCE STACK

Grant-Heavy. Low-Equity. BOO-Anchored. M4D Exposure Under \$500K.

TRANCHE	SOURCE & RATIONALE	ROM AMOUNT	COST	STRUCTURE
EPA Brownfields Assessment + Cleanup	Peninsula brownfield history. Funds Phase I/II ESA + remediation. DNREC environmental review is condition precedent for any DE construction.	\$500K–\$1.5M	0% grant	Non-dilutive
DEDO Strategic Fund + Brownfield Assistance	Delaware Economic Development Office. Manufacturing job creation in Wilmington EJ corridor is core DEDO mandate. Sean Park / OED is the introduction pathway already in M4D relationship.	\$500K–\$2M	0% grant	Non-dilutive
EDA Economic Adjustment Assistance	Wilmington qualifies as economically distressed. Manufacturing + clean tech job creation + brownfield remediation all EAA-eligible. Chester PA application establishes M4CT track record.	\$1M–\$3M	0% grant	Non-dilutive
Opportunity Zone Equity	Peninsula parcels likely within Wilmington OZ boundary. OZ equity investors defer capital gains. Attracts private co-investors alongside grant stack.	\$1M–\$3M	OZ equity	Some dilution — structure carefully
CDFI — Manufacturing Loan (Merge4Ward)	Artisans' Bank (DE-based), Reinvestment Fund, or NCALL Research. Manufacturing facility + equipment note at below-market rate. Ground lease as collateral support.	\$2M–\$4M	4–6%	15–20 yr term
Supplier Vendor Financing (Merge4Ward)	CLT manufacturers (TerraLam, Structurlam, Mercer) sometimes carry equipment/inventory notes on anchor customer relationships.	\$500K–\$1.5M	Negotiated	Secured on equipment

TRANCHE	SOURCE & RATIONALE	ROM AMOUNT	COST	STRUCTURE
	Negotiate as part of supply agreement.			
AD/RNG Operator Co-Investment (M4CT DE)	AD/RNG development partner brings operator equity + construction financing. M4CT contributes site control via ground lease. Developer fee + equity participation negotiated in development agreement.	\$6M–\$12M operator	Operator equity	No M4CT CapEx
Klean/KBR BOO — Tire Facility	Klean/KBR builds, owns, operates tire BOO on 26-052 southern tip parcel under sublease from M4CT. Zero capital from M4D regardless of outcome.	\$8M–\$15M operator	BOO operator	No M4CT CapEx
M4D / Merge4Ward / M4CT Equity	Principals' contribution. Minimized by grant stack. Ground lease eliminates land acquisition. Target: under \$500K combined across all campus entities.	<\$500K combined	Own capital	Recovered from dev fee
TOTAL CAMPUS — All Sources	Merge4Ward facility + AD/RNG + Tire BOO combined. M4D/M4CT combined capital exposure below \$500K.	\$19.5M–\$41.5M	Blended 2–3%	M4D exp: <\$500K

ROM ±30–40%. DEDO and EDA amounts depend on job creation commitments and competitive application quality. OZ equity requires qualified fund or investor — securities counsel required. CDFI loan availability depends on lender underwriting. AD/RNG operator terms subject to development agreement negotiation. Supplier vendor financing is subject to commercial negotiation with individual suppliers.

SECTION 5 — CHESTER + WILMINGTON: THE TWO-NODE MID-ATLANTIC STRATEGY

Each Node Strengthens the Other. Neither Depends on the Other.

Chester PA and Wilmington DE are not competing priorities. They are mutually reinforcing nodes in a coordinated Mid-Atlantic clean industrial and affordable housing strategy. Chester AD/RNG track record supports Wilmington CDFI underwriting. Merge4Ward Wilmington panels supply MaxWellness Chester construction. Chester grant awards establish M4CT credibility for Wilmington EDA application.

COMPONENT	CHESTER PA	WILMINGTON DE
AD/RNG Technology	Chester: Primary anchor. Bioenergy Devco / RNG Energy Solutions. 100–200 ton/day. Delaware River access for product distribution.	Wilmington: Second node. Same technology, same operator network. Port of Wilmington logistics for food waste intake and digestate offtake.
Tire BOO	Chester: 3–5 acre Reworld waterfront parcel. Klean/KBR TyreNova. Zero M4CT capital.	Wilmington: 26-052 southern tip — three-sided river frontage gives product barge offtake advantage. Klean/KBR second US deployment.
Construction Supply	Chester MaxWellness: Merge4Ward Wilmington panels shipped downriver. Lower cost than any competing panel supplier.	Merge4Ward Wilmington: Forward Building System anchor node for DE/PA/NJ/MD market. Ships to Chester, Baltimore, Camden.
Grant Track Record	Chester EPA EJ + EDA EAA first applications. First award cycle establishes M4CT credibility with federal agencies.	Wilmington EDA EAA second application — cites completed Chester grant cycle. Higher scoring due to proven delivery.
City Relationships	Chester: Mayor Stefan Roots. DCIDA (ROFR on Reworld site). Trevor leads relationship.	Wilmington: Sean Park / OED (existing M4D relationship). Mayor John Carney. Craig leads OED conversation.
CDFI Underwriting	Chester AD/RNG: tipping contracts + D3 RINs = first underwriting package. CDFI pre-qual initiated immediately.	Wilmington AD/RNG: Chester operating data (12–24 months) becomes second-application underwriting input. Stronger deal.
Combined Platform Revenue	Chester: \$7.35M–\$11.4M gross/yr (AD/RNG + MaxWellness + Tire BOO).	Wilmington: \$4.85M–\$8.8M gross/yr (Merge4Ward + AD/RNG + Tire BOO).

THE COMPOUNDING ADVANTAGE TREVOR SHOULD ARTICULATE TO BOTH CITIES

Chester and Wilmington are not two cities competing for M4D's attention. They are two anchor cities in a coordinated Mid-Atlantic clean industrial and affordable housing strategy. **Mayor Roots gets a clean tech anchor and a community health campus. Mayor John Carney gets a manufacturing node that supplies affordable housing panels to Delaware projects while co-locating a clean tech revenue generator.** Every Forward Home building in Delaware will use panels manufactured on the Wilmington waterfront — from a City-owned ground-leased site — creating a closed loop of Delaware-sourced, Delaware-manufactured, Delaware-delivered community housing.

SECTION 6 — IMMEDIATE ACTIONS — WILMINGTON CAMPUS

Four Actions. Two Cities. One Coordinated Timeline.

NOW — Q3 2026 OED Parcel Meeting Sean Park — existing relationship Present Hay Road Peninsula aerial + ForwardTech campus concept to Sean Park at Wilmington OED. Request: (1) parcel ownership confirmation on 26-044/26-045/26-052 series, (2) City's interest in master development agreement, (3) intro to Mayor John Carney. Bring Chester brief as proof of parallel momentum. Confirm rail spur status with CSX/NS before design.	Q3 2026 Supplier Pre-Qualification Ground lease as credential — before signing Initiate conversations with TerraLam, Structurlam, Acme Panel, and Aercon/Xella using the pending Wilmington ground lease as anchor credential. Do not finalize terms before lease is signed — but establish relationships and get indicative pricing that reflects committed-site buyer status. Also contact Bioenergy Devco re: Wilmington AD/RNG site feasibility.	Q3–Q4 2026 Ground Lease Negotiation + BOO Parallel DE transactional counsel required Engage DE transactional counsel for master development agreement. Simultaneously initiate Klean/KBR BOO site feasibility on 26-052 southern tip. DNREC pre-application meeting for AD/RNG facility. Right of First Refusal on adjacent parcels must be a ground lease term. File EPA Brownfields Assessment grant on peninsula.	Q4 2026–2027 Capital Stack Assembly Grant applications + CDFI pre-qual File DEDO Strategic Fund pre-application. File EDA EAA pre-application citing Chester as parallel project. Engage Artisans' Bank and Reinvestment Fund for Merge4Ward CDFI manufacturing loan pre-qualification. Identify OZ investors for campus equity component. OZ attorney engagement required before fund formation.
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All ROM investment, revenue, and savings figures are preliminary estimates with ±30–40% confidence range. Not guaranteed, not funded commitments, and not GMP. Produced for concept-level conversation and to frame the scope of M4Di predevelopment engagement. Revenue figures are gross before operating expenses. AD/RNG revenue depends on D3 RIN pricing, tipping fee contracts, and pipeline interconnection agreement terms — all require negotiated execution before financial close. Ground lease, BOO sublease, and supplier exclusivity arrangements require legal review. Grant capital is subject to competitive application and lender underwriting. Pennsylvania regulatory compliance requires PA-licensed counsel. DNREC environmental review and DEDO approvals are conditions precedent to any DE construction commitment.